Software Requirements Specification

Dynamics Queue Admin — dispatch-queue management console, version 1.0.0

Document	**SRS-1.0.0** — Software Requirements Specification
Application	Dynamics Queue Admin — the dispatch-queue (loading-place) management console of the Dynamics freight marketplace. Web application, product version 1.0.0.
Asset type	Web Application
Asset code	**ASSET-DYN-QA-001**
Standard	INSA Secure Website Management Standard, Version 1.0, 2014 EC
Production host	`queue.dynamicsroute.tech`
Backend API	`https://app.dynamicsroute.tech` (production) / `https://dev.dynamicsroute.tech` (development), reached at `<VITE_API_BASE_URL>/api`. The backend (`transportBackEndNative`) is a separate system of record.
Version specified	1.0.0
Scope	This web console only. The Dynamics backend, the driver mobile application, the shipper application and the operations/administrator dashboards are separate systems; their interfaces to this console are specified in §5 and §6.
Related documents	System & Architecture Design Document (SDD), Technical Design / Low-Level Design (TDD/LLD), End User Guide, Security Documentation; gap analysis `queue-admin-gap-analysis.md`; INSA standard content `src/data/insaStandard.ts`.
Submitting entity	**TBD**
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Prepared by	**TBD**
Reviewed by	**TBD**

Date of submission	**TBD**

Status of this document

This specification describes the console *as implemented* in the submitted release. Every requirement listed here is present in the shipped web application; no forward-looking features are specified. Where a capability is deliberately absent or dependent on an unimplemented backend service, that is stated honestly in the requirement or its note (§7.5). Fields shown in **amber** are administrative placeholders that the submitting entity completes before submission. Each requirement carries a

Source:

line pointing at the repository file that implements it.

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How to read this document

§3 contains the numbered functional requirements (prefix `FR-`), grouped by application area, each with a priority and a *Source* citation to the file that implements it. §4 contains the non-functional requirements (`NFR-`). §7 defines the queue, bidding and status business rules that most functional requirements refer to, and states plainly which decisions are made by the console and which are enforced by the backend (`transportBackEndNative`) as the system of record. Requirement identifiers are stable and are referenced by the SDD and by the test evidence, so a reviewer can trace a requirement to its design element and to its implementation.

> **Client-authority note**
>
> Dynamics Queue Admin is a client of the Dynamics backend. Authentication state, queue ordering, geofence decisions, bidding gate state, bid ranking and offer acceptance are **authoritative on the server**. The console renders that state and issues requests; it holds no independent authority over marketplace business data (§7.4).

1. Introduction

1.1 Purpose

This document specifies the requirements of Dynamics Queue Admin 1.0.0, the single-page web console used to operate a *loading place* (a Queue Organization) on the Dynamics freight marketplace. It is written for the submitting entity's own development and test baseline and for external review, and forms part of the submission package for the Information Security Authority's website review against the INSA Secure Website Management Standard (Version 1.0, 2014 EC).

1.2 Scope

Dynamics Queue Admin provides three jobs: (a) **loading-place management** — creating and administering a Queue Organization profile, its coordinates and optional check-in radius, its approval status, and its members (roles 11 and 12); (b) the **FIFO driver queue** — proving driver presence by GPS check-in and placing arrivals into a first-in-first-out line per vehicle type, then dispatching, overriding, removing and auditing entries; and (c) **bidding on the shipper's behalf** — approving or closing bidding per order and accepting a driver's bid on the shipper's behalf using explicit identifiers. It also covers a dashboard over all organizations, derived reporting, real-time synchronization, a security-compliance dashboard, localization (English and Amharic) and dark mode.

Out of scope: the Dynamics backend (`transportBackEndNative`), which is the system of record; the driver mobile application; the shipper application; and the operations and administrator dashboards. These are separate systems with separate requirements documents. All business logic of record — queue ordering, the check-in geofence, the bidding gate, journey status and offer acceptance — is held and enforced by the backend; this console is a client of that record and holds no authority over it (§7.4).

1.3 Definitions and acronyms

Term	Meaning
Queue Organization	A "loading place" tenant of the console (customs, factory, cement, depot, mine, farm, port, other) with coordinates, an optional check-in radius, an approval status and a queue-enabled flag.
Queue Org Admin	Backend `roleId: 11`. Administers the loading place: profile, membership and approval of bidding per order.
Queue Dispatcher	Backend `roleId: 12`. Runs the FIFO queue: manual check-in, dispatch, override, remove and entry history.
Admin / Super Admin	Backend `roleId: 3` / `roleId: 6`. Approve loading places and oversee all organizations. Short session (24 hours).
FIFO queue	First-in-first-out driver waiting line per vehicle type, keyed by the server-issued `queueNumber`.
Check-in	Proving driver presence at the loading place. The authoritative geofence decision for a GPS check-in is made by the backend; the console computes a Haversine distance for on-screen proximity cues.
targetedShipperUserUUID	An optional field on a queue entry that reserves a queued driver for one shipper; carried through dispatch as the shipper target.

Term	Meaning
Bidding	Drivers offer a price on an order; the console lists the authoritative bids cheapest-first and the operator accepts one on the shipper's behalf.
Bid	A driver's offer on an order (`driverBidUniqueId`, `bidAmount`, `bidStatus`), ranked by the backend.
Journey status	A backend status id (0-20) describing an order's lifecycle; only journey-status-1 orders are eligible for dispatch / bidding.
OTP	One-time password delivered by SMS; the sole authentication factor.
JWT	JSON Web Token issued after OTP verification and held in an httpOnly, SameSite cookie set server-side; never exposed to JavaScript.
Socket.IO	The real-time channel carrying the console's live queue and organization events.
RTK Query	Server-state data-fetching and caching layer; a single base `api` extended by feature endpoint modules.
Membership	A user's active registration under a Queue Organization with role 11 or 12; membership is the permission.
FR / NFR	Functional requirement / non-functional requirement identifier used in §3 and §4.

1.4 References

- `src/utils/constant.ts` — single source of truth for REST endpoint paths.
- `src/lib/redux/api/*.ts` — endpoint modules and RTK Query contract.
- `src/lib/redux/api/base.ts` — base query, credentials, 401-to-logout.
- `src/lib/socket.ts` — Socket.IO client, subscriptions and debounced invalidation.
- `src/routes/AppRoutes.tsx` — route table and lazy loading.
- `src/components/auth/ProtectedRoute.tsx`, `src/components/auth/RoleGuard.tsx` — guard stack.
- `src/types/queue.ts`, `src/lib/redux/api/types.ts` — domain types.
- `src/data/insaStandard.ts` — INSA standard content for the compliance dashboard.
- `package.json`, `vite.config.ts`, `.env.example` — build and environment model.
- `docs/queue-admin-gap-analysis.md` — known gaps referenced in §7.5.

1.5 Overview

§2 describes the product perspective, user classes, operating environment, constraints and dependencies. §3 enumerates the functional requirements in eleven families (FR-AUTH, FR-REG, FR-ORG, FR-MEM, FR-QUE, FR-BID, FR-ORD, FR-RPT, FR-NOTIF, FR-SET, FR-SECP). §4 enumerates non-functional requirements in seven families (NFR-SEC, NFR-PERF, NFR-USA, NFR-COMP, NFR-REL, NFR-MNT, NFR-PRIV). §5 specifies external interfaces (user, software, hardware, communications). §6 specifies data requirements. §7 states the business rules, the queue and journey status models, the client/backend authority boundary and the declared limitations. §8 gives requirements traceability and document control.

2. Overall description

2.1 Product perspective

Dynamics Queue Admin is one console in the Dynamics freight marketplace, a multi-application product. The marketplace backend (`transportBackEndNative`) is the system of record and exposes REST (`/api`) and Socket.IO interfaces. Drivers use a mobile application to check in, bid and update journey status; shippers use a separate application to create freight requests. This console is the dispatch side: it operates the physical loading place that drivers queue at and that shippers' requests are matched to. It is a single-page React application served as a static site at `queue.dynamicsroute.tech`.

2.2 User classes and characteristics

User class	Role id	Characteristics and primary tasks
Queue Org Admin	`11`	Primary administrator of a loading place. Creates and edits the organization profile (coordinates, radius, type), manages members, and approves or closes bidding per order. Long-lived session (365 days).
Queue Dispatcher	12	Runs the daily queue. Manually checks drivers in, dispatches queued entries to requests, overrides position or removes entries with a reason, and reviews entry history. Long-lived session (365 days).
Admin	`3`	Platform administrator. Approves, rejects or suspends loading places and oversees all organizations. Short session (24 hours).
Super Admin	6	Platform super-administrator. Same oversight and approval powers as Admin across all organizations. Short session (24 hours).

All four classes are authenticated by phone number plus OTP and gated by the guard stack (`ProtectedRoute` then `RoleGuard`, which admits roles `{11, 12, 3, 6}`). Client-side role checks are a usability boundary only; the backend re-checks role and membership on every request.

2.3 Operating environment

- **Runtime:** a modern evergreen browser executing a Vite-built React 19 SPA (ES modules, dynamic import for route chunks).
- **Network:** HTTPS to `queue.dynamicsroute.tech`; REST and Socket.IO to the backend host. Session cookie is httpOnly/SameSite and requires `credentials: "include"`.
- **Storage:** `localStorage` for non-sensitive preferences only (`queueadmin:user` profile, `app_dark_mode`, `app_language`, persisted `selectedOrgId`); no token is ever stored.
- **Optional external service:** the Photon geocoder for order-place search (`VITE_PHOTON_URL`), with a built-in Ethiopian-cities fallback.

2.4 Design and implementation constraints

- **Backend authority:** the console must not compute or invent authoritative identifiers (queue numbers, bid ids, membership ids); it must use values returned by the backend and send them back explicitly (see FR-BID and §7.4).

- **Cookie session:** a JWT must not be placed in JavaScript-accessible storage; `getToken()` returns `null` by design.
- **Role-resolved sign-in:** the login and OTP endpoints must be called without a `roleId` so that the backend resolves the account's own role and a role-12 dispatcher can sign in.
- **Single API source of truth:** endpoint paths must live in `src/utils/constant.ts`; feature modules must not hard-code paths.
- **Hand-written CSS only:** no utility framework; design tokens live in `src/index.css` with a `src/dark.css` dark theme.

2.5 Assumptions and dependencies

- The backend enforces authentication, role and membership on every request.
- The backend issues and validates the session cookie over HTTPS with SameSite attributes.
- The backend exposes the Socket.IO event fan-out under the `queueOrgAdmin` identity for queue staff and `admin` for platform admins.
- The backend performs geofence evaluation for GPS check-in using the organization's coordinates and `checkinRadiusKm`.
- The Photon geocoder is optional; when unavailable, place search falls back to a built-in city list.
- The hosting environment terminates TLS and, where the INSA standard requires it, applies HTTP security headers (no CSP is configured in this repository; see §7.5).

2.6 Stakeholders and external actors

Actor / system	Interest
Queue Org Admin (11)	Operate the loading place; control bidding; manage members.
Queue Dispatcher (12)	Keep the physical queue accurate and fair (FIFO).
Admin (3) / Super Admin (6)	Approve and oversee loading places across the marketplace.
Driver / vehicle owner	Fair queue position; visible bid outcomes via the driver application.
Shipper	Receive a dispatched driver or an accepted bid for their request.
Backend REST API	System of record; authoritative business rules and identifiers.
Backend Socket.IO	Real-time queue and organization event fan-out.
Photon geocoder (optional)	External place search for order origin/destination.
INSA reviewer	Assess the console against the Secure Website Management Standard.

3. Functional requirements

Priority: **M** must have (the console is not releasable without it), **S** should have. All requirements below are implemented in version 1.0.0 unless a note states otherwise. Each requirement cites its implementing repository file under *Source*.

3.1 Authentication, session and registration (FR-AUTH)

ID	Requirement	Pri
FR-AUTH-01	The console shall register a Queue Org Admin account with the backend by calling `POST /api/user/createUser` with the full name, phone number, email and status fields, registering backend role 11, before OTP verification. Source: `src/lib/redux/api/authEndpoints.ts`, `src/utils/constant.ts`	M
FR-AUTH-02	The console shall sign a returning user in with a phone number via `POST /api/user/loginUser`, which dispatches an OTP to the user's phone. This call shall deliberately send **no `roleId`** so the backend resolves the account's own role and a role-12 dispatcher can sign in. Source: `src/utils/constant.ts:16`, `src/lib/redux/api/authEndpoints.ts`	M
FR-AUTH-03	The console shall verify the OTP via `POST /api/user/verifyUserByOTP`, likewise sending no `roleId`, and shall persist only the non-secret `userData` profile returned by the backend. Source: `src/lib/redux/api/authEndpoints.ts` (verifyOtp `onQueryStarted`)	M
FR-AUTH-04	The OTP screen shall render six single-digit inputs with automatic advance on entry and back-navigation on delete, and shall enable the verify action only when all six digits are present. Source: `src/components/auth/OtpDigitInputs.tsx`, `src/components/auth/VerifyOtp.tsx`	M
FR-AUTH-05	The OTP screen shall offer a resend action governed by a countdown timer that re-issues the login OTP when it elapses, and shall surface OTP errors through a toast. Source: `src/hooks/useOtpVerification.ts`, `src/components/auth/VerifyOtp.tsx`	S
FR-AUTH-06	On successful verification the backend shall set the session as an httpOnly, SameSite cookie (JWT); the console shall send it on every request using `credentials: "include"` and shall never read or store a token in JavaScript (`getToken()` returns `null`). Source: `src/lib/redux/api/base.ts`, `src/lib/api.ts`, `src/lib/auth.ts:84`	M
FR-AUTH-07	The console shall persist only the non-sensitive user profile under `localStorage["queueadmin:user"]`, and on first read shall migrate any legacy `queueadmin:auth` record that still held a JWT by dropping the token and keeping only `userData`. Source: `src/lib/auth.ts`	M
FR-AUTH-08	The Redux `authSlice` shall initialize from the stored profile (`hydrated: true`) so the UI can render optimistically across reloads without a blocking session round-trip. Source: `src/lib/redux/slices/authSlice.ts`	M
FR-AUTH-09	When an authenticated profile is present the console shall connect the Socket.IO client (carrying the phone number); when the profile is cleared it shall disconnect the socket. Source: `src/context/AuthContext.tsx`, `src/lib/socket.ts`	M

ID	Requirement	Pri
FR-AUTH-10	The console shall log out by making a best-effort `POST /api/user/logout` to clear the server cookie and shall always clear local authentication state (profile and Redux auth) whether or not that call succeeds. Source: `src/context/AuthContext.tsx`, `src/lib/auth.ts`	M
FR-AUTH-11	The RTK Query base query shall detect any `401` response and dispatch a global `logout()`, clearing the auth slice everywhere at once; there is no refresh-token flow, so reauth equals logout. Source: `src/lib/redux/api/base.ts` (`baseQueryWithReauth`)	M
FR-AUTH-12	The legacy axios instance shall use `withCredentials` and a `401` response interceptor that clears authentication and routes the user back to `/login`. Source: `src/lib/api.ts`	M
FR-AUTH-13	A Socket.IO `connect_error` whose message indicates `unauthorized` or `token` shall disconnect the socket and end the session, and shall stop further reconnect attempts. Source: `src/lib/socket.ts`	M
FR-AUTH-14	Route access shall be enforced by a two-stage guard stack: `ProtectedRoute` shall require an authenticated user and, by default, at least one Queue Organization (redirecting to `/setup-org` when none exists, with `/setup-org` itself passing `requireOrg={false}`); the nested `RoleGuard` shall admit only roles `{11, 12, 3, 6}` and otherwise redirect to `/login`. Source: `src/components/auth/ProtectedRoute.tsx`, `src/components/auth/RoleGuard.tsx`, `src/routes/AppRoutes.tsx`	M
FR-AUTH-15	While no session exists, the console shall render public routes only (`/login`, `/register`) and shall not issue non-public backend traffic; the public endpoints `loginUser`, `createUser`, `verifyUserByOTP` and `logout` remain reachable. Source: `src/routes/AppRoutes.tsx`, `src/components/auth/ProtectedRoute.tsx`	M

3.2 Registration and onboarding (FR-REG)

ID	Requirement	Pri
FR-REG-01	The registration screen shall collect full name and phone number (required) and email (optional), driven by a declarative field configuration with `roleId: 11` as the default account role. Source: `src/components/auth/Register.tsx`, `src/lib/authConfig.ts`	M
FR-REG-02	Phone input shall be normalized to Ethiopian E.164 format (`+251` plus nine digits), stripping a leading `0` or `251`, and grouped for readability as the user types. Source: `src/lib/authConfig.ts`, `src/utils/phoneFormatter.ts`, `src/components/ui/PhoneNumberInput.tsx`	M
FR-REG-03	Registration and login inputs shall be validated with `react-hook-form` plus `zod` schemas that constrain length, format and required fields, and that sanitize text and email against injected markup and control characters. Source: `src/schemas/queue.ts`, `src/components/auth/Login.tsx`	M
FR-REG-04	Registration submission shall call `POST /api/user/createUser` and, on success, transition the auth flow to OTP verification. Source: `src/components/auth/Register.tsx`, `src/lib/redux/api/authEndpoints.ts`	M

ID	Requirement	Pri
FR-REG-05	When an authenticated user has no Queue Organization, the console shall route them to the onboarding screen (`/setup-org`) to create their first loading place. Source: `src/components/auth/ProtectedRoute.tsx`, `src/components/organization/SetupOrganization.tsx`	M
FR-REG-06	The onboarding screen shall submit a new Queue Organization with name, type, coordinates and optional check-in radius, and the backend shall create it with `approvalStatus: "pending"`. Source: `src/components/organization/SetupOrganization.tsx`, `src/lib/redux/api/queueOrganizationEndpoints.ts`	M
FR-REG-07	Organization types shall be restricted to the enum `customs | factory | cement | depot | mine | farm | port | other`. Source: `src/types/queue.ts` (`QUEUE_ORG_TYPES`), `src/schemas/queue.ts`	M
FR-REG-08	After onboarding, a pending organization shall display an approval banner explaining that platform approval is required before the queue is enabled. Source: `src/components/dashboard/OrgApprovalBanner.tsx`, `src/components/queue-org-manage/StatusBadge.tsx`	S

3.3 Loading-place (organization) management (FR-ORG)

ID	Requirement	Pri
FR-ORG-01	The console shall create a loading place by calling `POST /api/queueOrganization` with the profile fields, producing a `pending` organization. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/utils/constant.ts:25`	M
FR-ORG-02	The console shall list the organizations available to the signed-in user via `GET /api/queueOrganization` and render them in a dashboard list with pagination and a toolbar. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/hooks/useOrgList.ts`, `src/components/dashboard/OrgListToolbar.tsx`	M
FR-ORG-03	The console shall fetch a single organization via `GET /api/queueOrganization/:id` for the management page. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/pages/queue-org-manage/QueueOrgManagePage.tsx`	M
FR-ORG-04	The console shall update the organization profile via `PATCH /api/queueOrganization/:id`, including name, type, coordinates and check-in radius. Source: `src/components/queue-org-manage/OrgProfileForm.tsx`, `src/lib/redux/api/queueOrganizationEndpoints.ts`	M
FR-ORG-05	The console shall capture latitude and longitude for the loading place and validate them against legal coordinate ranges before submission. Source: `src/schemas/queue.ts`, `src/components/queue/order-modal/CoordinatesRow.tsx`	M
FR-ORG-06	The console shall support an optional `checkinRadiusKm` (integer), presenting a default of 15 km when unset, and shall treat a value of 0 as disabling the radius gate. The authoritative geofence decision remains server-side. Source: `src/components/queue-org-manage/OrgProfileForm.tsx`, `src/utils/formatters.ts` (`calculateDistanceKm`)	M

ID	Requirement	Pri
FR-ORG-07	The console shall compute great-circle (Haversine) distance between two coordinate pairs for on-screen proximity cues only, and shall not treat that client figure as the geofence authority. Source: `src/utils/formatters.ts`, `src/components/orders/bids/BidRow.tsx`	M
FR-ORG-08	The console shall delete (soft-delete) a loading place via `DELETE /api/queueOrganization/:id` after confirmation. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/components/queue/ConfirmCancel.tsx`	S
FR-ORG-09	The console shall drive the approval workflow via `PATCH /api/queueOrganization/:id/approve` to approve, reject or suspend a loading place. Approval toggles `queueEnabled`; the backend is authoritative over which roles may approve. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/components/queue-org-manage/AdminApprovalActions.tsx`	M
FR-ORG-10	The console shall allow selecting an active organization, persisted as `selectedOrgId`, and shall scope the queue and orders views to that selection. Source: `src/hooks/useActiveOrg.ts`, `src/store/queueAdminStore.ts`, `src/lib/redux/slices/queueSlice.ts`	M
FR-ORG-11	The dashboard shall show organization status (pending / approved / rejected / suspended) with a status badge and, for the active organization, a summary view with queue metrics. Source: `src/components/dashboard/ActiveOrgView.tsx`, `src/components/dashboard/OrgStatusTable.tsx`, `src/components/queue-org-manage/StatusBadge.tsx`	M
FR-ORG-12	The management page shall render a queue details modal that lists the active organization's queue entries and per-entry metrics. Source: `src/components/queue/OrgQueueDetailsModal.tsx`, `src/components/queue/org-details/OrgQueueMetricsGrid.tsx`	S

3.4 Membership management (FR-MEM)

ID	Requirement	Pri
FR-MEM-01	The console shall list an organization's members via `GET /api/queueOrganization/:id/members`, showing user, role (11/12) and active status. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/components/queue-org-manage/MembersTable.tsx`	M
FR-MEM-02	The console shall add a member via `POST /api/queueOrganization/:id/members`, sending the `userUniqueId` in the request **body** (not the path). Source: `src/utils/constant.ts:30`, `src/lib/redux/api/queueOrganizationEndpoints.ts`	M
FR-MEM-03	The console shall deactivate a membership via `PATCH /api/queueOrganization/:id/members/:membershipId/deactivate`, revoking access immediately. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/components/queue-org-manage/MembersTable.tsx`	M

ID	Requirement	Pri
FR-MEM-04	The console shall reactivate a membership via `PATCH /api/queueOrganization/:id/members/:membershipId/reactivate`. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`	S
FR-MEM-05	The console shall remove a membership via `DELETE /api/queueOrganization/:id/members/:membershipId` after confirmation. Source: `src/lib/redux/api/queueOrganizationEndpoints.ts`, `src/components/queue/ConfirmCancel.tsx`	S
FR-MEM-06	Because membership is the permission, the members view shall reflect the active flag and role exactly as returned by the backend, and the backend shall require an active membership row on every queue request. Source: `src/components/queue-org-manage/MembersTable.tsx`, `src/types/queue.ts`	M
FR-MEM-07	Member rows shall expose deactivate/reactivate/remove actions; the console shall invalidate the members query after each lifecycle mutation. Source: `src/components/queue-org-manage/MembersTable.tsx`, `src/lib/redux/api/queueOrganizationEndpoints.ts`	M
FR-MEM-08	The console shall present members' roles using the shared labels for role 11 (Queue Org Admin) and role 12 (Queue Dispatcher). Source: `src/components/queue-org-manage/MembersTable.tsx`, `src/types/queue.ts:34`	S

3. Functional requirements (continued)

3.5 FIFO driver queue (FR-QUE)

ID	Requirement	Pri
FR-QUE-01	The console shall fetch the queue snapshot for an organization and date via `GET /api/queue/status`, returning entries grouped by vehicle type. Source: `src/lib/redux/api/driverQueueEndpoints.ts`, `src/utils/constant.ts:38`	M
FR-QUE-02	The queue board shall offer a date selector and refetch the snapshot when the selected date changes, using the organization's real (not display) identifier. Source: `src/components/queue/QueueBoardHeader.tsx`, `src/components/queue/QueueBoard.tsx`	M
FR-QUE-03	The console shall check a vehicle-driver into the queue via `POST /api/queue/manualCheckin`; the backend returns the server-issued `queueNumber` that fixes the entry's FIFO position. Source: `src/lib/redux/api/driverQueueEndpoints.ts`, `src/components/queue/CheckinModal.tsx`	M
FR-QUE-04	Manual check-in shall be offered only where the organization is approved and the queue is enabled; the backend remains authoritative over whether a check-in is accepted. Source: `src/components/queue/CheckinModal.tsx`, `src/components/dashboard/OrgApprovalBanner.tsx`	M
FR-QUE-05	The check-in form shall search the driver directory via `GET /api/queue/driverDirectory` (queue-staff only), returning active driver/vehicle assignments. Source: `src/lib/redux/api/vehicleEndpoints.ts`, `src/components/queue/checkin/CheckinDriverSearch.tsx`	M
FR-QUE-06	The check-in search shall merge the driver directory with the current queue so that already-queued drivers are visible and not duplicated. Source: `src/components/queue/checkin/useCheckinData.ts`	M
FR-QUE-07	Check-in input (driver, vehicle, position) shall be validated with a `zod` schema before submission, and the selected driver and vehicle identifiers shall be sent explicitly. Source: `src/schemas/queue.ts`, `src/components/queue/CheckinModal.tsx`	M
FR-QUE-08	The check-in screen shall provide a position section and show a Haversine proximity cue relative to the organization, labelled as an aid rather than the geofence decision. Source: `src/components/queue/checkin/CheckinPositionSection.tsx`, `src/utils/formatters.ts`	S
FR-QUE-09	The console shall order queue entries by the server-issued `queueNumber` (first-in-first-out) within each vehicle type, and shall not reorder entries locally. Source: `src/components/queue/table/QueueDesktopTable.tsx`, `src/utils/journeyStatus.ts`	M
FR-QUE-10	The queue board shall present entries grouped into per-vehicle-type cards (`QueueVehicleTypeCard`) for at-a-glance monitoring. Source: `src/components/queue/QueueBoard.tsx`, `src/components/queue/QueueVehicleTypeCard.tsx`	M
FR-QUE-11	The board shall provide an all-drivers aggregate view across vehicle types. Source: `src/components/queue/QueueAllDriversCard.tsx`	S

ID	Requirement	Pri
FR-QUE-12	The console shall render the queue as a desktop table with the entry's driver, vehicle type, status and FIFO number, and per-row actions. Source: `src/components/queue/table/QueueDesktopTable.tsx`, `src/components/queue/QueueTable.tsx`	M
FR-QUE-13	The console shall render a dedicated mobile queue view (cards) alongside the desktop table. Source: `src/components/queue/table/QueueMobileTable.tsx`, `src/components/queue/QueueCard.tsx`	S
FR-QUE-14	The console shall dispatch a queued entry to a shipper request via `POST /api/queue/dispatch`, sending the entry and the target request. Source: `src/lib/redux/api/driverQueueEndpoints.ts`, `src/components/queue/DispatchModal.tsx`	M
FR-QUE-15	The dispatch order selector shall offer only orders eligible for dispatch (journey status 1), using the shared journey-status mapping. Source: `src/components/queue/dispatch/OrderSelectDropdown.tsx`, `src/utils/journeyStatus.ts`	M
FR-QUE-16	The dispatch modal shall show a top-card summary and an order summary for the pending dispatch before confirmation. Source: `src/components/queue/dispatch/DispatchTopCards.tsx`, `src/components/queue/dispatch/DispatchOrderSummary.tsx`	S
FR-QUE-17	A queue entry's optional `targetedShipperUserUUID` shall be displayed and carried through dispatch as the reserved shipper target. Source: `src/components/queue/table/buildShipperRequestDetail.ts`, `src/components/queue/shipper-requests/ShipperRequestOverview.tsx`	M
FR-QUE-18	The console shall override a queue entry's position via `PATCH /api/queue/entry/:queueUniqueId/override`, requiring a reason; the change is audited in the backend queue audit trail. Source: `src/components/queue/OverrideModal.tsx`, `src/lib/redux/api/driverQueueEndpoints.ts`	M
FR-QUE-19	The console shall remove a queue entry via `DELETE /api/queue/entry/:queueUniqueId` after a confirmation step; the removal is audited. Source: `src/components/queue/ConfirmCancel.tsx`, `src/lib/redux/api/driverQueueEndpoints.ts`	M
FR-QUE-20	The console shall render the entry change history from `GET /api/queue/entry/:queueUniqueId/history` (column, old value, performed by, performed at), and shall display entry statuses (`waiting | offered | loaded | removed | completed`) and the refusal count as returned by the backend. Source: `src/components/queue/EntryHistoryModal.tsx`, `src/types/queue.ts`	M

3.6 Bidding on the shipper's behalf (FR-BID)

ID	Requirement	Pri
FR-BID-01	The console shall approve or close bidding per order via `POST /api/queue/bidding/approve`, sending the `shipperRequestUniqueIds` list and the `approved` flag. Source: `src/lib/redux/api/driverQueueEndpoints.ts`, `src/components/orders/bids/BiddingBoardToggle.tsx`	M
FR-BID-02	The bidding gate shall be applied **per order**, not per batch, so each order carries its own `isBiddingApproved` state. Source: `src/components/orders/bids/BiddingBoardToggle.tsx`, `src/components/orders/OrdersTable.tsx`	M
FR-BID-03	The bidding toggle shall be presented only to the Queue Org Admin role (11); the backend independently enforces approval permission. Source: `src/components/orders/bids/BiddingBoardToggle.tsx`, `src/components/auth/RoleGuard.tsx`	M
FR-BID-04	The console shall fetch the authoritative driver bids for an order via `GET /api/queue/bidding/order/:shipperRequestUniqueId/bids`, presented cheapest-first as ranked by the backend. Source: `src/lib/redux/api/driverQueueEndpoints.ts`, `src/components/orders/DriverBidsModal.tsx`	M
FR-BID-05	The bids view shall provide a toolbar for refreshing and filtering the bid list. Source: `src/components/orders/bids/BidsToolbar.tsx`, `src/components/orders/bids/useDriverBidsFilter.ts`	S
FR-BID-06	The console shall normalize heterogeneous bid payloads into a single UI shape before rendering. Source: `src/components/orders/bids/mapDriverBids.ts`	M
FR-BID-07	The bids view shall show an order summary card (origin, destination, vehicle type, requested quantity) alongside the bids. Source: `src/components/orders/bids/OrderSummaryCard.tsx`, `src/components/orders/bids/BidOfferComparison.tsx`	S
FR-BID-08	Individual bid rows shall display the driver, amount, distance and originating request, and shall expose accept actions per bid. Source: `src/components/orders/bids/BidRow.tsx`, `src/components/orders/bids/BidActionButtons.tsx`	M
FR-BID-09	The console shall accept a driver's offer via `PUT /api/shipper/acceptDriverOffer`, sending the explicit `shipperRequestUniqueId` and `driverRequestUniqueId` (plus an optional `journeyDecisionUniqueId`). Source: `src/components/orders/bids/useAcceptDriverBid.ts`, `src/lib/redux/api/driverQueueEndpoints.ts`	M
FR-BID-10	If a required identifier is missing or invalid, acceptance shall **hard-fail** with a visible error; the console shall never reconstruct or guess an identifier (no fuzzy match). Source: `src/components/orders/bids/orderIdLookup.ts`, `src/components/orders/bids/useAcceptDriverBid.ts`	M

ID	Requirement	Pri
FR-BID-11	After a successful acceptance the console shall invalidate the affected order, bid and queue tags so all dependent views refetch authoritative state, and shall surface the outcome through a toast. Source: `src/components/orders/bids/useAcceptDriverBid.ts`, `src/lib/redux/api/driverQueueEndpoints.ts`	M
FR-BID-12	When no bids are available the console shall render an explicit empty state rather than a blank list. Source: `src/components/orders/bids/BidEmptyState.tsx`	S

3.7 Orders and shipper requests (FR-ORD)

ID	Requirement	Pri
FR-ORD-01	The console shall list shipper requests (all or single, scoped to the organization) via `GET /api/user/getShipperRequest4allOrSingleUser`. Source: `src/lib/redux/api/shipperRequestEndpoints.ts`, `src/pages/orders/OrdersPage.tsx`	M
FR-ORD-02	The console shall list company-target batches via `GET /api/shipperRequestBatch` with `requestMode=company_target` and `includeBids=true`, rendering master rows with sub-rows. Source: `src/lib/redux/api/shipperRequestEndpoints.ts`, `src/components/orders/BatchMasterRow.tsx`	M
FR-ORD-03	The console shall create a shipper request or batch order via a modal that calls `POST /api/shipperRequest/createRequest` with origin, destination, vehicle type, quantity, dates and request mode. Source: `src/components/queue/CreateOrderModal.tsx`, `src/lib/redux/api/shipperRequestEndpoints.ts`	M
FR-ORD-04	When the requested number of vehicles exceeds nine in individual mode, the modal shall automatically switch the request to `company_target` and open it for bidding by default. Source: `src/components/queue/CreateOrderModal.tsx` (auto-switch effect)	M
FR-ORD-05	If the operator switches back to individual mode while more than nine vehicles are set, the modal shall reset the quantity to one. Source: `src/components/queue/CreateOrderModal.tsx` (`handleRequestModeChange`)	S
FR-ORD-06	Orders shall be rendered across a desktop table, cards list and dedicated mobile views with status chips and pagination. Source: `src/components/orders/OrdersTable.tsx`, `src/components/orders/OrdersCardsList.tsx`, `src/components/orders/OrdersMobileCards.tsx`	M
FR-ORD-07	The console shall provide edit and delete modals for orders, with confirmation before deletion. Source: `src/components/orders/OrdersEditModal.tsx`, `src/components/orders/OrdersDeleteModal.tsx`	S
FR-ORD-08	Order creation shall offer a vehicle-type select backed by `GET /api/admin/vehicleTypes` (with a built-in fallback list) and shall show the derived vehicle capacity. Source: `src/components/queue/order-modal/VehicleTypeSelect.tsx`, `src/utils/vehicleType.ts`, `src/lib/redux/api/vehicleEndpoints.ts`	M

3.8 Reporting (FR-RPT)

ID	Requirement	Pri
FR-RPT-01	The reports page shall render KPI cards summarizing organizations, queue and request counts. Source: `src/pages/reports/ReportsPage.tsx`, `src/components/reports/KpiCards.tsx`	M
FR-RPT-02	The reports page shall render a donut chart of the queue/order distribution. Source: `src/components/reports/DonutChart.tsx`	S
FR-RPT-03	The reports page shall render a monthly bar chart derived from request dates. Source: `src/components/reports/MonthlyBarChart.tsx`	S
FR-RPT-04	The reports page shall list a per-organization report with per-row drill-down. Source: `src/components/reports/OrgReportsList.tsx`, `src/components/reports/OrgReportRow.tsx`	M
FR-RPT-05	Report metrics shall be computed client-side from the fetched organization, queue and shipper-request lists; this is a declared limitation for scale (§7.5). Source: `src/hooks/useReportsMetrics.ts`	M
FR-RPT-06	The dashboard shall derive its headline counts client-side from the same fetched lists (limit 100), consistent with the reports computation. Source: `src/pages/dashboard/QueueDashboardPage.tsx`, `src/hooks/useOrgList.ts`	S

3. Functional requirements (continued)

3.9 Real-time and notifications (FR-NOTIF)

ID	Requirement	Pri
FR-NOTIF-01	The console shall open a Socket.IO connection with `withCredentials: true` and an `auth` payload of `{ user, phoneNumber }`. Source: `src/lib/socket.ts`	M
FR-NOTIF-02	The socket identity shall be derived from the account role: roles 3 and 6 map to `admin`; queue staff (roles 11 and 12) share the `queueOrgAdmin` identity used by the backend fan-out. Source: `src/lib/socket.ts`	M
FR-NOTIF-03	The console shall subscribe to a Queue Organization room via `queue:subscribe`, keyed by `queueOrganizationUniqueId` and an optional `queueDate`, with reference counting so the room is unsubscribed only when the last consumer leaves. Source: `src/lib/socket.ts`, `src/hooks/useQueueSocket.ts`	M
FR-NOTIF-04	The console shall unsubscribe via `queue:unsubscribe` when a component no longer needs the room. Source: `src/lib/socket.ts`	M
FR-NOTIF-05	The socket shall reconnect with up to five attempts, a randomized delay between 1.5 s and 5 s, and a 20 s timeout, and shall re-subscribe active rooms on (re)connect. Source: `src/lib/socket.ts`	M
FR-NOTIF-06	The console shall treat a `queue:subscribed` acknowledgement as confirmation that the room is live and set the shared `socketConnected` flag. Source: `src/lib/socket.ts`, `src/store/queueAdminStore.ts`	M
FR-NOTIF-07	Inbound queue events shall be parsed for a `messageTypes` value; acknowledgements and ping/pong frames shall be ignored, so no spurious refetches occur. Source: `src/lib/socket.ts`	M
FR-NOTIF-08	Organization events shall debounce 100 ms and then invalidate the `QueueOrganizations` tag. Source: `src/lib/socket.ts`	M
FR-NOTIF-09	All other queue events shall debounce 250 ms and then invalidate `QueueStatus`, `DriverQueue` and `ShipperRequests`, triggering authoritative refetches rather than per-event cache patching. Source: `src/lib/socket.ts`, `src/hooks/useQueueSocket.ts`	M
FR-NOTIF-10	The console shall surface operation outcomes and errors to the user through `sonner` toasts rendered at the application root. Source: `src/App.tsx`, `src/utils/parseError.ts`	M

3.10 Settings, profile and appearance (FR-SET)

ID	Requirement	Pri
FR-SET-01	The console shall provide a settings page composed of profile, preferences and account sections, with dedicated mobile settings views. Source: src/pages/settings/SettingsPage.tsx, src/components/settings/MobileSettingsView.tsx	M
FR-SET-02	The console shall provide a light/dark theme toggle that applies the dark class to the document root and persists the preference under app_dark_mode. Source: src/contexts/ThemeContext.tsx, src/dark.css	M
FR-SET-03	The console shall provide an English/Amharic language selector that persists the preference under app_language. Source: src/i18n/index.ts, src/components/ui/LanguageSelector.tsx	M
FR-SET-04	Internationalization shall be configured with escapeValue: false because React performs output escaping, and components shall use inline fallback strings. Source: src/i18n/index.ts, src/i18n/en.ts, src/i18n/am.ts	M
FR-SET-05	The profile section shall display the signed-in user's stored profile; because no profile-update endpoint exists, the edit controls shall be disabled rather than faked. Source: src/pages/settings/SettingsPage.tsx, src/components/settings/DesktopProfileSection.tsx	S
FR-SET-06	Push, email and two-factor controls shall be disabled with a visible explanation, since no notification-preference or 2FA backend endpoint exists. Source: src/pages/settings/SettingsPage.tsx:101, src/components/settings/DesktopPreferences.tsx	S
FR-SET-07	The settings page shall provide a danger zone with a delete-account confirmation modal. Source: src/components/settings/DangerZone.tsx, src/components/settings/DeleteAccountModal.tsx	S
FR-SET-08	The console shall expose mobile account and preferences sections in addition to the desktop layout. Source: src/components/settings/MobileAccountSection.tsx, src/components/settings/MobilePreferencesSection.tsx	S

3.11 Security compliance dashboard (FR-SECP)

ID	Requirement	Pri
FR-SECP-01	The console shall render a security-compliance dashboard from the INSA Secure Website Management Standard content held in src/data/insaStandard.ts. Source: src/pages/security/SecurityPage.tsx, src/data/insaStandard.ts	M
FR-SECP-02	The dashboard shall present five focus areas (requirement gathering, design, implementation, hosting, operation & management) with thirty-five requirements. Source: src/data/insaStandard.ts, src/components/security/FocusAreaAccordion.tsx	M

ID	Requirement	Pri
FR-SECP-03	The dashboard shall present the eight common vulnerabilities (Annex A) in a table. Source: `src/data/insaStandard.ts`, `src/components/security/VulnerabilityTable.tsx`	M
FR-SECP-04	The dashboard shall present a 102-item security-test checklist (Annex B) that a reviewer can mark pass/fail, with the results persisted locally. Source: `src/components/security/SecurityChecklist.tsx`, `src/hooks/useSecurityChecklist.ts`	M
FR-SECP-05	The dashboard shall compute and display a compliance score and summary cards from the marked checklist items. Source: `src/utils/securityChecklist.ts`, `src/components/security/SecurityScoreCards.tsx`	S

4. Non-functional requirements

Non-functional requirements are grouped by family. Each cites its implementing repository file. Where a quality depends on the hosting environment rather than this code asset, that is stated explicitly.

4.1 Security (NFR-SEC)

ID	Requirement	Pri
NFR-SEC-01	Authentication shall rely on an httpOnly, SameSite session cookie set by the backend; no JWT shall be readable from JavaScript, and `getToken()` shall return `null`. Source: `src/lib/auth.ts`, `src/lib/redux/api/base.ts`	M
NFR-SEC-02	Every backend request shall send the cookie with `credentials: "include"` and shall not set an `Authorization` header. Source: `src/lib/redux/api/base.ts`, `src/lib/api.ts`	M
NFR-SEC-03	Any `401` shall trigger a single global logout so an expired session cannot leave the app in a half-authenticated state. Source: `src/lib/redux/api/base.ts`, `src/lib/api.ts`	M
NFR-SEC-04	A socket handshake rejected as unauthorized shall end the session and stop reconnect attempts. Source: `src/lib/socket.ts`	M
NFR-SEC-05	Client route guards shall be a usability boundary only; every request shall be re-authorized by the backend against role and active membership. Source: `src/components/auth/ProtectedRoute.tsx`, `src/components/auth/RoleGuard.tsx`	M
NFR-SEC-06	All forms shall validate and sanitize input with `zod`, stripping HTML tags, control characters and collapsing whitespace before submission. Source: `src/schemas/queue.ts`	M
NFR-SEC-07	JavaScript-accessible storage shall hold only non-sensitive values (`queueadmin:user`, `app_dark_mode`, `app_language`, persisted `selectedOrgId`), and legacy token records shall be purged on read. Source: `src/lib/auth.ts`, `src/store/queueAdminStore.ts`	M
NFR-SEC-08	Production builds shall disable source maps. Source: `vite.config.ts` (`sourcemap`)	M
NFR-SEC-09	**Honest limitation:** this repository does not configure a Content Security Policy or other HTTP security headers; header hardening is a server-level responsibility recorded in §7.5. Source: `index.html` (no meta CSP), `docs/queue-admin-gap-analysis.md`	S
NFR-SEC-10	Error handling shall mask internal details by normalizing server and client errors into human-readable messages, mapping status codes to safe fallbacks. Source: `src/utils/parseError.ts`	M

4.2 Performance and efficiency (NFR-PERF)

ID	Requirement	Pri
NFR-PERF-01	Every route shall be lazily loaded so each page ships in its own dynamic-import chunk behind a suspense fallback. Source: `src/routes/AppRoutes.tsx`, `src/main.tsx`	M
NFR-PERF-02	The production build shall split vendor libraries into separate manual chunks (vendor, redux, icons) to improve caching and first load. Source: `vite.config.ts`	S
NFR-PERF-03	Cached server state shall be retained for 300 s (`keepUnusedDataFor: 300`) to avoid redundant fetches. Source: `src/lib/redux/api/base.ts`	M
NFR-PERF-04	The client shall disable refetch-on-focus and refetch-on-reconnect, relying on explicit invalidation and the realtime channel instead. Source: `src/lib/redux/api/base.ts`	M
NFR-PERF-05	Realtime events shall be debounced and coalesced (100 ms / 250 ms) so a burst of events produces a single refetch per affected tag. Source: `src/lib/socket.ts`	M
NFR-PERF-06	Distance comparisons shall be computed in constant time per coordinate pair, and list computations shall be bounded by the API list limits (100 items). Source: `src/utils/formatters.ts`, `src/hooks/useOrgList.ts`	S

4.3 Usability and accessibility (NFR-USA)

ID	Requirement	Pri
NFR-USA-01	The console shall provide responsive desktop and mobile layouts across the queue, orders, reports, settings and dashboard. Source: `src/index.css`, `src/components/common/MobileHeader.tsx`	M
NFR-USA-02	Interactive controls and modals shall expose ARIA roles/labels and keyboard handling, including a modal accessibility hook. Source: `src/hooks/useModalA11y.ts`, `src/components/ui/Modal.tsx`	M
NFR-USA-03	The interface shall be available in English and Amharic. Source: `src/i18n/en.ts`, `src/i18n/am.ts`, `src/i18n/index.ts`	M
NFR-USA-04	The console shall offer a persisted light/dark theme. Source: `src/contexts/ThemeContext.tsx`	S
NFR-USA-05	All user-initiated operations shall give immediate feedback (loading state plus toast). Source: `src/components/ui/SubmitButton.tsx`, `src/App.tsx`	M
NFR-USA-06	Data views shall render explicit loading and empty states rather than blank regions. Source: `src/components/dashboard/OrgListState.tsx`, `src/components/orders/bids/BidEmptyState.tsx`	S

4.4 Compatibility and portability (NFR-COMP)

ID	Requirement	Pri
NFR-COMP-01	The console shall run on modern evergreen browsers that support ES modules and the Fetch API. Source: `vite.config.ts`, `package.json`	M
NFR-COMP-02	The static build shall be served with an SPA fallback so deep links resolve, with the in-app `*` route redirecting to `/dashboard`. Source: `src/routes/AppRoutes.tsx`, `vite.config.ts`	M
NFR-COMP-03	The API base URL shall be resolved from environment configuration at build time (`VITE_API_BASE_URL`, falling back to `VITE_API_URL` then `/api`), so the same source targets dev and production without code change. Source: `src/utils/baseUrl.ts`, `.env.example`	M

4.5 Reliability and availability (NFR-REL)

ID	Requirement	Pri
NFR-REL-01	The console shall render optimistically from the stored profile so a reload does not block on a session round-trip. Source: `src/lib/redux/slices/authSlice.ts`, `src/context/AuthContext.tsx`	M
NFR-REL-02	Only a real `401` shall end the session; a timeout, an offline browser, a `4xx` other than 401, or a `5xx` shall not terminate the session. Source: `src/lib/redux/api/base.ts`, `src/utils/parseError.ts`	M
NFR-REL-03	The realtime channel shall recover from transient disconnects by reconnecting and re-subscribing without a page reload. Source: `src/lib/socket.ts`	M
NFR-REL-04	After every mutation the console shall invalidate the affected tags so that local state reconciles with the backend even if a prior optimistic view was stale. Source: `src/lib/redux/api/*.ts`	M
NFR-REL-05	**Honest limitation:** the console does not declare a React error boundary; an uncaught render error is not contained in-app. This is recorded in §7.5 as a known gap. Source: `src/main.tsx`, `src/App.tsx`	S

4.6 Maintainability (NFR-MNT)

ID	Requirement	Pri
NFR-MNT-01	All REST endpoint paths shall live in a single source of truth, `src/utils/constant.ts`, and constants shall be maintained only when they point at a real backend route with at least one consumer. Source: `src/utils/constant.ts`	M

ID	Requirement	Pri
NFR-MNT-02	The API layer shall be strongly typed; endpoint modules inject into one base `api` and expose generated hooks from a shared barrel. Source: `src/lib/redux/api/index.ts`, `src/lib/redux/api/types.ts`	M
NFR-MNT-03	Error handling shall be centralized so no feature re-implements message extraction. Source: `src/utils/parseError.ts`	M
NFR-MNT-04	Features shall be modular by area (queue, orders, organization, reports, security, settings) with co-located components and styles. Source: `src/components/*`, `src/pages/*`	S
NFR-MNT-05	The project shall maintain an automated test suite (98 tests across 7 files covering queue mutations, orders mapping, journey status, phone formatting, INSA standard and the security checklist). Source: `src/__tests__/*.test.ts`	M
NFR-MNT-06	The visual system shall be token-based hand-written CSS with a dark theme, avoiding a utility-framework dependency. Source: `src/index.css`, `src/dark.css`	S

4.7 Privacy and data protection (NFR-PRIV)

ID	Requirement	Pri
NFR-PRIV-01	The console shall collect only the data needed to operate the loading place and queue (profile, organization, membership, queue and order data); it shall embed no advertising or third-party analytics. Source: `package.json`, `index.html`	M
NFR-PRIV-02	Only non-sensitive data shall persist in the browser; personal data at rest shall remain on the backend. Source: `src/lib/auth.ts`, `src/store/queueAdminStore.ts`	M
NFR-PRIV-03	No authentication token or credential shall be written to `localStorage` or `sessionStorage`. Source: `src/lib/auth.ts`	M
NFR-PRIV-04	Personal data shall be processed by the backend as data controller; the console shall display it only within an authorized session and shall clear it on logout. Source: `src/context/AuthContext.tsx`, `src/lib/auth.ts`	M

5. External interface requirements

5.1 User interfaces

The console is a single-page application. The screens below are reached through the route table and are lazy-loaded. All operator-facing screens are rendered in the console's own design system (hand-written CSS); there is no third-party UI kit.

Screen	Route	Purpose
Sign in	`/login`	Phone number entry; dispatch OTP.
OTP verification	`/login` (step)	Six-digit OTP entry and resend.
Register	`/register`	Create a Queue Org Admin account.
Setup organization	`/setup-org`	Onboard the first loading place.
Dashboard	`/dashboard`	Organization list, status and active-org summary.
Organization management	`/orgs/:queueOrganizationUniqueId`	Profile, approval actions and members.
Orders	`/orders`, `/orgs/:queueOrganizationUniqueId/orders`	Shipper requests, batches and bids.
Reports	`/reports`	KPIs, charts and per-org reports.
Security compliance	`/security`	INSA self-assessment dashboard.
Settings	`/settings`	Profile, theme, language and account.
Queue board	(within dashboard / org views)	FIFO queue, check-in, dispatch, override, history.
Fallback	`/`, `/organizations`, `*`	Redirect to `/dashboard`.

5.2 Software interfaces

Endpoint paths are centralized in `src/utils/constant.ts`; the API base resolves to `<VITE_API_BASE_URL>/api`. The frontend-callable surface is:

Method + path (`/api` prefix)	Purpose	Module
`POST /user/loginUser`	Request login OTP for a phone number (no `roleId`).	auth
`POST /user/verifyUserByOTP`	Verify OTP; backend sets the session cookie (no `roleId`).	auth
`POST /user/createUser`	Register a queue-org admin account (role 11).	auth
`POST /user/logout`	Clear the server session cookie.	auth

Method + path (`/api` prefix)	Purpose	Module
`GET /queueOrganization`	List queue organizations for the user.	queue organization
`POST /queueOrganization`	Create a loading place (`approvalStatus: pending`).	queue organization
`GET /queueOrganization/:id`	Get one organization.	queue organization
`PATCH /queueOrganization/:id`	Update the organization profile.	queue organization
`DELETE /queueOrganization/:id`	Soft-delete the organization.	queue organization
`PATCH /queueOrganization/:id/approve`	Approve / reject / suspend; toggles `queueEnabled` .	queue organization
`GET /queueOrganization/:id/members`	List members (role 11/12, active flag).	queue organization
`POST /queueOrganization/:id/members`	Add a member (`userUniqueId` in body).	queue organization
`PATCH /queueOrganization/:id/members/:membershipId/deactivate`	Deactivate membership (revokes access immediately).	queue organization
`PATCH /queueOrganization/:id/members/:membershipId/reactivate`	Reactivate membership.	queue organization
`DELETE /queueOrganization/:id/members/:membershipId`	Remove membership.	queue organization
`GET /queue/status`	Queue snapshot by vehicle type for a date.	driver queue
`POST /queue/manualCheckin`	Check a vehicle-driver in; returns `queueNumber` .	driver queue
`POST /queue/dispatch`	Dispatch a queued entry to a shipper request.	driver queue
`DELETE /queue/entry/:queueUniqueId`	Remove a queue entry (audited).	driver queue
`PATCH /queue/entry/:queueUniqueId/override`	Override position with a reason (audited).	driver queue
`GET /queue/entry/:queueUniqueId/history`	Entry change history.	driver queue

Method + path (`/api` prefix)	Purpose	Module
`GET /queue/driverDirectory`	Search active driver/vehicle assignments (queue staff only).	vehicles
`POST /queue/bidding/approve`	Open/close bidding per order (`shipperRequestUniqueIds`, `approved`).	driver queue / bids
`GET /queue/bidding/order/:shipperRequestUniqueId/bids`	Authoritative driver bids for an order (cheapest-first).	driver queue / bids
`POST /shipperRequest/createRequest`	Create a shipper request / batch order.	shipper requests
`GET /user/getShipperRequest4allOrSingleUser`	List shipper requests (all/single, org-scoped).	shipper requests
`GET /shipperRequestBatch`	List batches (`company_target`, `includeBids`).	shipper requests
`PUT /shipper/acceptDriverOffer`	Accept a driver bid with explicit ids.	driver queue / bids
`GET /admin/vehicleTypes`	Reference vehicle types (falls back to built-in defaults).	vehicles

5.2.1 Authentication and session contract

- The primary session is an httpOnly cookie set by the backend at `POST /api/user/verifyUserByOTP`. The client never reads or stores a JWT (`getToken()` returns `null`).
- Every request sends the cookie with `credentials: "include"`; no `Authorization` header is set.
- There is no refresh-token flow: any `401` dispatches a global logout in the RTK Query base query, and the axios interceptor clears auth and routes to `/login`.
- Session lifetime is role-based on the backend: roles 3 and 6 = 24 hours; roles 11, 12 and others = 365 days.

5.2.2 Socket.IO protocol

The client connects with `withCredentials` and an `auth` payload of `{ user, phoneNumber }`. The socket identity is derived from the account role: roles 3 and 6 map to `admin`; queue staff (roles 11 and 12) share a single `queueOrgAdmin` identity because the backend fan-out keys queue events under that identity for both roles. Subscriptions use `queue:subscribe` / `queue:unsubscribe` with a reference-counted room keyed by `queueOrganizationUniqueId` (and optional `queueDate`); rooms are re-subscribed on (re)connect. Inbound events are parsed for a `messageTypes` value; acknowledgements and ping/pong are ignored. Organization events invalidate `QueueOrganizations` after 100 ms; all other queue events invalidate `QueueStatus`, `DriverQueue` and `ShipperRequests` after 250 ms. A rejected handshake ends the session.

5.2.3 Error-handling contract

`src/utils/parseError.ts` normalizes axios errors, RTK Query errors, serialized errors and generic throws into one human-readable message, reading server `message` / `error` / `details` / `validation` fields and mapping HTTP status codes to fallbacks. `401` means "session expired — log in again" and triggers logout; `403` means "authenticated but not permitted" and is shown as a permission error without logging out.

5.2.4 Optional third-party service

Place search for order origin/destination uses the Photon geocoder (`VITE_PHOTON_URL`, default `https://photon.komoot.io/api/`) merged with a built-in list of Ethiopian cities and a local fallback. No map tiles or routing calls are made by this console.

5.3 Hardware interfaces

No dedicated hardware is required. The console needs a device with a modern browser and network access. An optional device geolocation capability is used only to compute an on-screen proximity cue; the authoritative check-in geofence is evaluated by the backend.

5.4 Communications interfaces

- HTTPS (TLS) to the SPA host and the backend REST API.
- WebSocket (Socket.IO) for realtime queue and organization events.
- Cookies: the httpOnly/SameSite session cookie, sent with `credentials: "include"`.
- SMS: OTP delivery, performed by the backend and the telecom provider.

6. Data requirements

6.1 Core entities and key attributes

Entity	Key attributes (as used by the console)
User	`userId`, `userUniqueId`, `fullName`, `phoneNumber`, `email`, `roleId` (11, 12, 3, 6).
QueueOrganization	`queueOrganizationUniqueId`, name, type enum (`customs | factory | cement | depot | mine | farm | port | other`), `latitude` / `longitude`, `checkinRadiusKm`, `approvalStatus` (`pending | approved | rejected | suspended`), `queueEnabled`, `approvedBy` / `approvedAt`.
QueueOrganizationMembership	`queueOrganizationMembershipUniqueId`, `userUniqueId`, `roleId` (11/12), `isActive`, `membershipStartDate`. Membership is the permission.
DriverQueue	`queueUniqueId`, `queueNumber`, `queueStatus` (`waiting | offered | loaded | removed | completed`), `queueRefusalCount`, `targetedShipperUserUUID`, `vehicleDriverUniqueId`, `shipperRequestUniqueId`, timestamps.
DriverQueueHistory	`historyUniqueId`, `columnName`, `oldValue`, `performedBy`, `performedAt`.
QueueAuditLog	Backend audit trail for queue mutations (check-in, dispatch, override, remove) including the override reason and acting user; surfaced in the entry history view.
ShipperRequest	`shipperRequestUniqueId`, `shipperRequestBatchUniqueId`, `requestMode` (`individual_target | company_target`), origin/destination, `shippingCost`, `journeyStatusId`, `queueOrganizationUniqueId`, `isBiddingApproved`.
ShipperRequestBatch	`batchUniqueId`, `totalVehicles`, `requestMode`, origin/destination, and aggregated bids (fetched with `includeBids=true`, `requestMode=company_target`).
DriverBid	`driverBidUniqueId`, `shipperRequestUniqueId`, `driverUserUniqueId`, `bidAmount`, `bidNotes`, `bidStatus` (`pending | selected | not_selected | withdrawn | expired`), `driverBidCreatedAt`.

6.2 Client state and storage

Store / storage	Keys / areas	Notes
RTK Query cache	Tags QueueOrganizations, QueueOrgMembers, QueueStatus, DriverQueue, DriverBids, DriverDirectory, Auth, VehicleTypes, ShipperRequests	`keepUnusedDataFor: 300`; tag invalidation; no focus/reconnect refetch.
Redux slices	`authSlice` (profile + hydration), `queueSlice` (`selectedOrgId`, `orgsCache`)	Registered in `src/lib/redux/store.ts`.
zustand store	`queueadmin:store` — persisted `selectedOrgId` plus transient `socketConnected`	Persisted via the zustand persist middleware.

Store / storage	Keys / areas	Notes
localStorage	`queueadmin:user` (non-secret profile), `app_dark_mode`, `app_language`	Never stores tokens; legacy `queueadmin:auth` token record is purged on read.
httpOnly cookie	Session cookie (backend-set)	Set at `POST /api/user/verifyUserByOTP`; sent with `credentials: "include"`.

6.3 Data flow

UI actions invoke RTK Query hooks/mutations or legacy axios services. The base query resolves the API prefix from `VITE_API_BASE_URL` / `VITE_API_URL` (normalized to end in `/api`) and sends the httpOnly cookie via `credentials: "include"`. Responses populate the RTK Query cache; socket events trigger debounced tag invalidation that refetches affected queries. Client state (auth profile, selected organization, theme, language, socket flag) lives in Redux slices, zustand, React Context and localStorage; no token is ever placed in JavaScript-accessible storage.

6.4 Data validation constraints

- Phone numbers are normalized to `+251` plus nine digits.
- Latitude is constrained to [-90, 90] and longitude to [-180, 180].
- Vehicle quantities are positive integers; greater-than-nine individual requests are coerced to batch (`company_target`).
- UUID and enum fields are validated against their allowed forms before submission.
- Text and email inputs are sanitized (tags, control characters and whitespace collapsed).

7. Business rules and status model

7.1 Queue business rules

ID	Rule	Authority
`BR-Q-01`	A driver's presence at the loading place is proven by GPS check-in; the geofence compares the driver's position against the organization's coordinates within `checkinRadiusKm` (default 15 km; 0 disables the gate).	Backend (server-authoritative); the console shows a Haversine proximity cue only.
`BR-Q-02`	Queue order is strictly first-in-first-out by the server-issued `queueNumber` within each vehicle type.	Backend (issues the number); the console preserves and displays it.
`BR-Q-03`	An entry may be positioned by an override, which requires a reason and is written to the backend audit trail.	Backend (audits); the console collects the reason.
`BR-Q-04`	An entry may be removed; removal is audited and reversible only through a new check-in.	Backend; the console requests removal after confirmation.
`BR-Q-05`	A queue entry may carry a `targetedShipperUserUUID` that reserves the driver for one shipper; it is carried through dispatch as the shipper target.	Backend; the console displays and forwards it.
`BR-Q-06`	Only an approved organization with the queue enabled may accept check-ins and dispatches.	Backend; the console hides the affordances where it can detect the state.

7.2 Bidding business rules

ID	Rule	Authority
`BR-B-01`	Bidding is gated per order, not per batch; each order carries its own approval flag.	Backend; the console toggles and displays per order.
`BR-B-02`	Only orders whose journey status is 1 are eligible for bidding / dispatch.	Backend; the console filters the selector.
`BR-B-03`	Bids are ranked and returned cheapest-first by the backend.	Backend; the console renders the order as returned.
`BR-B-04`	Accepting a bid requires the explicit `shipperRequestUniqueId` and `driverRequestUniqueId` (optional `journeyDecisionUniqueId`).	Backend; the console sends the exact ids.
`BR-B-05`	A missing or invalid identifier is a hard failure; identifiers are never reconstructed, guessed or fuzzy-matched.	Client rule enforced in the accept hook; backend rejects invalid ids.
`BR-B-06`	Only the Queue Org Admin role (11) is presented with bidding approval controls.	Backend enforces; the console gates the control.

7.3 Status models

Model	Values
Organization approval status	`pending` → `approved` / `rejected` / `suspended` (approval toggles `queueEnabled`)
Queue entry status	`waiting | offered | loaded | removed | completed`
Bid status	`pending | selected | not_selected | withdrawn | expired`
Request mode	`individual_target | company_target` (10+ vehicles is forced to batch)
Journey status	Backend status id 0-20 mapped to a name and label by `src/utils/journeyStatus.ts`; status 1 denotes an order awaiting a driver.
User role	`11` Queue Org Admin, `12` Queue Dispatcher, `3` Admin, `6` Super Admin

7.4 Client authority boundary

Concern	Console (client)	Backend (authority)
Route visibility	Guards hide and redirect unauthorized routes.	Re-checks role and active membership on every request.
Queue order	Displays entries in the order returned; never reorders.	Issues `queueNumber` and enforces FIFO.
Check-in geofence	Shows a Haversine distance cue.	Makes the authoritative GPS geofence decision.
Bidding gate	Toggle shown only to role 11; reflects server state.	Enforces approval permission and gate state per order.
Bid ranking and acceptance	Renders the cheapest-first list; sends explicit ids.	Ranks bids and validates the accepted offer.
Membership	Reflects the active membership row.	Requires an active membership row per request.

7.5 Declared limitations

The following are known, deliberate or deferred limitations of version 1.0.0, stated honestly for the reviewer. None of them is represented elsewhere in this document as a capability.

Item	Status	Detail / planned action
No refresh-token flow	NOTED	Any expiration ends the session (401 → logout); add refresh only if the backend offers it.
Role-dependent session lifetime	BY DESIGN	Roles 3/6 = 24 hours; roles 11/12 and others = 365 days, set by the backend.
Unused `@tanstack/react-query` dependency	OPEN	Nothing imports it; remove from `package.json`.

Item	Status	Detail / planned action
Settings profile / notifications / 2FA disabled	NOTED	No profile-update or notification/2FA backend; controls are disabled with a visible explanation rather than faked.
Dashboard counts computed client-side	NOTED	Metrics derive from fetched lists (limit 100) in the browser; move to server aggregates for scale.
Socket payloads trigger refetch only	BY DESIGN	Events invalidate tags; no per-event cache patching is applied.
Manual check-in merges directory with current queue	BY DESIGN	The picker merges the driver directory with the current queue to avoid duplicates.
Unreachable `/organizations` page	OPEN	The route redirects to `/dashboard`; the standalone page is not reachable and should be deleted or re-wired.
No CSP or HTTP security headers in this repo	DEFERRED	Server-level control outside this code asset; mitigate at the hosting / reverse-proxy layer.
No in-app error boundary	OPEN	Add a top-level React error boundary to contain render errors.
localStorage holds non-sensitive profile only	BY DESIGN	No token is stored; legacy token records are purged on read.

8. Requirements traceability and document control

8.1 Requirement traceability

Each requirement family maps to implementing repository elements, and is elaborated by the companion SDD. The table below records the primary mapping.

Family	Count	Primary implementation
FR-AUTH	15	`authEndpoints.ts`, `useOtpVerification.ts`, `AuthContext.tsx`, `lib/auth.ts`, `base.ts`, guards
FR-REG	8	`Register.tsx`, `authConfig.ts`, `SetupOrganization.tsx`, `schemas/queue.ts`
FR-ORG	12	`queueOrganizationEndpoints.ts`, `QueueOrgManagePage.tsx`, `OrgProfileForm.tsx`, dashboard components
FR-MEM	8	`queueOrganizationEndpoints.ts`, `MembersTable.tsx`
FR-QUE	20	`driverQueueEndpoints.ts`, `components/queue/*`, `journeyStatus.ts`
FR-BID	12	`components/orders/bids/*`, `DriverBidsModal.tsx`, `driverQueueEndpoints.ts`
FR-ORD	8	`OrdersPage.tsx`, `CreateOrderModal.tsx`, `shipperRequestEndpoints.ts`
FR-RPT	6	`ReportsPage.tsx`, `useReportsMetrics.ts`, `components/reports/*`
FR-NOTIF	10	`lib/socket.ts`, `useQueueSocket.ts`, `App.tsx`
FR-SET	8	`SettingsPage.tsx`, `ThemeContext.tsx`, `i18n/*`
FR-SECP	5	`SecurityPage.tsx`, `data/insaStandard.ts`, `securityChecklist.ts`
NFR-SEC / PERF / USA / COMP / REL / MNT / PRIV	40	`base.ts`, `parseError.ts`, `schemas/queue.ts`, `vite.config.ts`, `i18n/*`, `__tests__/*`

8.2 Test evidence

An automated test suite covers the highest-risk logic: queue mutations, order mapping, journey-status mapping, phone formatting, the INSA standard content and the security checklist. Tests are colocated under `src/__tests__/` and are run with the project test script. Requirements above cite the files exercised by those tests, giving a requirement-to-code-to-test chain.

8.3 Document control

Field	Value
Document identifier	SRS-1.0.0 — Dynamics Queue Admin

Field	Value
Asset code	ASSET-DYN-QA-001
Version	1.0.0
Status	As-built specification of the submitted release
Standard	INSA Secure Website Management Standard, Version 1.0, 2014 EC
Prepared by	**TBD**
Reviewed by	**TBD**
Approved by	**TBD**
Date of submission	**TBD**

> **Placeholders**
>
> **TBD** is used only where the value is strictly unknown (submitting entity, contact, prepared/reviewed/approved by, and dates). All technical details in this document reflect the actual repository state of `queadmin-frontend` version 1.0.0.